The Common-Reconstruction Rate–Distortion Function with Noisy Context

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Abstract

An encoder observes a jointly Gaussian pair (Y, V) and describes Y at rate R for a consumer that sees the description alone. A third party holds a noisy copy $S = V + U$ of the context V. Neither the consumer nor the third party decodes with S; instead S prices the description's conditional content $L = I(X; \hat{X} \mid S)$, the minimal ideal work, in units of $k_B T \ln 2$, that a reset mechanism retaining S must pay to erase the record. We characterize the minimal conditional content at consumer distortion D. The function has a closed form: $L(D) = \frac{1}{2} \log_2 g^*$, where g^* is the larger root of an explicit quadratic in the correlation ρ between read and context and the context noise τ^2, in any ambient dimension. It recovers the classical rate–distortion function, Gray's conditional rate–distortion function, and Steinberg's common-reconstruction function as degenerations, and it is not a functional of the (Y, S) joint law alone: two instances with the same (Y, S) law and different encoder access have different L. We give the exact rate–content region, whose Pareto frontier is a two-water-level system, and prove that a strict tradeoff between rate and content opens exactly when the context is misaligned with the read. We settle when the natural determinant floor is attained. For the doubly symmetric binary source with a binary symmetric context channel the function closes completely, with an elementary proof, through a tilt equation anchored at Gray's conditional function on one end and the marginal rate–distortion function on the other. Every closed form is verified by independent numerical certificates.

Index Terms

Rate–distortion theory, side information, common reconstruction, conditional mutual information, Landauer principle, Gaussian sources.

I. Introduction

A stored description of data has two prices. The first is its rate, the space the description occupies, and rate–distortion theory prices it against the fidelity a consumer requires. The second is paid at the end of the description's life: erasing a record irreversibly costs work, and when the mechanism performing the reset retains correlated information, the work is governed not by the record's entropy but by its conditional entropy given what the mechanism holds. The two prices are set by different parties. The consumer never sees the reset mechanism's information, and the reset mechanism never serves the consumer. This paper solves the source coding problem that this asymmetry creates, for Gaussian and binary sources, in closed form.

The setting is as follows. An encoder observes a jointly Gaussian pair (Y, V): Y is the variable a consumer cares about, V is context correlated with it ($\rho = \mathbb{E}[YV]$ after normalization). The encoder emits a description under a distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \le D$ that binds the unconditional error; the consumer decodes from the description alone. A third party holds $S = V + U$, a noisy observation of the context with noise variance τ^2, and possibly a stale one. The quantity of interest is $L(D) = \min I(X; \hat{X} \mid S)$ over test channels satisfying the distortion constraint and the Markov chain $\hat{X} - X - S$: the minimal conditional content of any admissible description, and, by the operational theorem of Section II, the minimal ideal erasure work per symbol in units of $k_B T \ln 2$.

Three classical objects sit at the corners of this problem, and the function we characterize is none of them. When the read and the context decouple ($\rho = 0$), L is Shannon's rate–distortion function. When the third party's copy is clean ($\tau^2 \to 0$), L is Gray's conditional rate–distortion function [3]. When read and context merge ($\rho^2 \to 1$),

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L is Steinberg's common-reconstruction function [7] at the induced correlation. Between the corners the function is new, and the distinction is not cosmetic: we exhibit two instances with the same (Y, S) joint law whose values of L differ (Corollary 8). The encoder's access to the clean context V, which neither the consumer nor the third party shares, is load-bearing. To our knowledge no vector-Gaussian common-reconstruction-type function has been published under any formulation; the scalar Gaussian case is due to Steinberg [7] and, for general jointly Gaussian side information, Lapidoth, Malär, and Wigger [8].

The main results are these. Theorem 6 reduces the problem in any ambient dimension to the pair (ρ, τ^2). Theorem **??** gives the closed form $L(D) = \frac{1}{2}\log_2 g^*$ with g^* the larger root of an explicit quadratic, together with the achieving channel. Theorem **??** characterizes the exact region of achievable rate–content pairs; its Pareto frontier is traced by a two-water-level stationarity system, and Corollary **??** shows the frontier is nondegenerate, so that rate and content genuinely trade off, exactly when $\rho^2 \in (0,1)$ and $\tau^2 > 0$. Theorem 9 settles when the natural determinant floor is attained, correcting a plausible conjecture: the attainment condition involves the encoder-accessible conditional covariance, not the third party's. Theorem **??** extends the frontier to vector context. Theorem 10 closes the binary case completely: for a doubly symmetric binary source with a binary symmetric context channel, an elementary and self-contained proof yields the function through a tilt equation, anchored at Gray's binary conditional function at one end and the marginal rate–distortion function at the other.

II. Problem Formulation and the Operational Theorem

All logarithms are base two. Random variables have finite alphabets unless a Gaussian specialization is stated. Let $(X, S) \sim p_{XS}$, and let (X^n, S^n) be i.i.d. copies. The encoder observes X^n but not S^n. The consumer and its decoder also do not observe S^n. The reset mechanism alone has access to S^n, which remains unchanged by the reset.

A distortion measure is a bounded function

$$d : \mathcal{X} \times \widehat{\mathcal{X}} \longrightarrow [0, d_{\max}]. \tag{1}$$

An $(n, \mathcal{M}_n)$ observation code consists of an encoder and decoder

$$f_n : \mathcal{X}^n \to \mathcal{M}_n, \qquad g_n : \mathcal{M}_n \to \widehat{\mathcal{X}}^n. \tag{2}$$

We write $M = f_n(X^n)$ and $\widehat{X}^n = g_n(M)$. Randomized encoders do not improve the region below, but allowing them in the converse is harmless.

The three operational quantities are

$$R_n = \frac{1}{n}\log|\mathcal{M}_n|, \tag{3}$$

$$D_n = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\, d(X_i, \widehat{X}_i), \tag{4}$$

$$L_n = \frac{1}{n}H(M \mid S^n). \tag{5}$$

The dimensionless quantity L_n is the *Landauer content* of the memory relative to the retained side information. Under an ideal conditional reset, the average work satisfies

$$W_n \;\geq\; nk_{\mathrm{B}}T\ln 2\, L_n. \tag{6}$$

In the classical asymptotic regime, conditional erasure protocols make this bound tight up to sublinear terms [14], [15]. We therefore state the coding theorem in bits per source symbol and restore physical units by multiplying by $k_{\mathrm{B}}T\ln 2$. The bound (6) prices only the logically irreversible reset of the volatile index M; it does not count the work of acquiring X^n, driving a channel, correcting errors, or finishing in finite time [12], [13]. The codebook, wiring, and any permanent lookup tables are treated as part of the apparatus rather than a per-cycle record.

Definition 1 (Achievability). *A triple (R, L, D) is achievable if there is a sequence of observation codes such that*

$$\limsup_{n\to\infty} R_n \leq R, \qquad \limsup_{n\to\infty} L_n \leq L, \qquad \limsup_{n\to\infty} D_n \leq D. \tag{7}$$

The closure of the achievable set at distortion D is denoted $\mathcal{RW}(D)$.

For a test channel $p(\hat{x} \mid x)$, the Markov chain $\widehat{X} - X - S$ holds because side information is unavailable when the observation is formed. Define

$$\mathcal{T}(D) = \left\{ p(\hat{x} \mid x) : \mathbb{E}d(X, \widehat{X}) \leq D \right\}. \tag{8}$$

Lemma 2 (Convexity in the test channel)**.** *For fixed p_{XS}, both coordinates $I(X; \widehat{X})$ and $I(X; \widehat{X} \mid S)$ are convex functions of the test channel $q(\hat{x} \mid x)$.*

Proof. Convexity of $I(X; \widehat{X})$ in the channel at fixed input is classical. For the conditional coordinate, the Markov chain $\widehat{X} - X - S$ gives $p(\hat{x} \mid x, s) = q(\hat{x} \mid x)$, so

$$I(X; \widehat{X} \mid S) = \sum_{s} p(s)\, I_{p(x|s)}(q), \tag{9}$$

a nonnegative combination of mutual informations, each with fixed input distribution $p(x \mid s)$ and the common channel q; each term is convex in q, hence so is the sum. We note that the decomposition $I(X; \widehat{X} \mid S) = I(X; \widehat{X}) - I(S; \widehat{X})$ does *not* yield this conclusion: $I(S; \widehat{X})$ is itself convex in q (the induced channel $p(\hat{x} \mid s) = \sum_x p(x \mid s) q(\hat{x} \mid x)$ is linear in q), so the difference is a priori indefinite. □

Theorem 3 (Rate–work–distortion region)**.** *For a discrete memoryless source (X, S) and bounded distortion d,*

$$\boxed{\mathcal{RW}(D) = \bigcup_{p(\hat{x}|x) \in \mathcal{T}(D)} \left\{ (R, L) : R \geq I(X; \widehat{X}),\ L \geq I(X; \widehat{X} \mid S) \right\}.} \tag{10}$$

The physical reset-work coordinate is $W/n = k_{\mathrm{B}} T \ln 2\, L$ in the ideal model. By Lemma 2 the union is already convex, and it is closed by compactness of $\mathcal{T}(D)$ and continuity of both coordinates; no time sharing is required, and every boundary point is attained by a single test channel.

Proof. We first prove the converse. Let a code produce M and $\widehat{X}^n$. Since $\widehat{X}^n$ is a function of M,

$$\begin{aligned} \log |\mathcal{M}_n| &\geq H(M) \geq I(X^n; M) \geq I(X^n; \widehat{X}^n) \\ &\geq \sum_{i=1}^{n} I(X_i; \widehat{X}_i), \end{aligned} \tag{11}$$

where the last step follows from memorylessness and conditioning reducing entropy (Appendix A). Likewise,

$$\begin{aligned} H(M \mid S^n) &\geq I(X^n; M \mid S^n) \geq I(X^n; \widehat{X}^n \mid S^n) \\ &\geq \sum_{i=1}^{n} I(X_i; \widehat{X}_i \mid S_i). \end{aligned} \tag{12}$$

Each induced per-letter channel $p(\hat{x}_i \mid x_i)$ satisfies the Markov constraint $\widehat{X}_i - X_i - S_i$: $\widehat{X}_i$ is a function of $(X_i, X_{\neq i})$ and S_i is independent of $X_{\neq i}$ given X_i for i.i.d. pairs. Let $\bar{q}(\hat{x} \mid x) = \frac{1}{n} \sum_{i=1}^{n} p(\hat{x}_i = \hat{x} \mid X_i = x)$ be the mixture channel; the constraint survives the mixture because $p(s \mid x)$ is common across i. Its distortion is D_n, so $\bar{q} \in \mathcal{T}(D_n)$, and by Lemma 2,

$$R_n \ \geq\ \frac{1}{n} \sum_{i} I(X_i; \widehat{X}_i) \ \geq\ I_{\bar{q}}(X; \widehat{X}), \qquad L_n \ \geq\ \frac{1}{n} \sum_{i} I(X_i; \widehat{X}_i \mid S_i) \ \geq\ I_{\bar{q}}(X; \widehat{X} \mid S),$$

which places (R_n, L_n) in the (already convex and closed) union of (10) at distortion D_n.

For achievability, fix $p(\hat{x} \mid x) \in \mathcal{T}(D)$ and $\epsilon > 0$. Generate a standard rate–distortion codebook containing $2^{n(I(X;\widehat{X})+2\epsilon)}$ independent $\hat{x}^n$ codewords. A covering encoder selects a codeword jointly typical with x^n and stores its index M. The consumer reconstructs that codeword from M alone. Standard typicality arguments give rate $I(X; \widehat{X}) + 2\epsilon$ and distortion at most $D + o(1)$.

To bound the reset work, randomly partition the codeword indices into $2^{n(I(X;\widehat{X}|S)+3\epsilon)}$ bins, and let $B = b(M)$ be the bin index. Because $\widehat{X} - X - S$,

$$I(X; \widehat{X} \mid S) = I(X; \widehat{X}) - I(S; \widehat{X}). \tag{13}$$

The usual Wyner–Ziv packing argument [4] therefore gives a decoder that recovers M from (B, S^n) with probability of error tending to zero. This decoder is a proof device; the consumer still receives no side information. By Fano's inequality,

$$\begin{aligned} H(M \mid S^n) &\le H(B) + H(M \mid B, S^n) \\ &\le n\big(I(X; \widehat{X} \mid S) + 3\epsilon\big) + o(n). \end{aligned} \tag{14}$$

Thus the same stored index has ordinary description rate $I(X; \widehat{X})$ and conditional Landauer content $I(X; \widehat{X} \mid S)$. Since the union in (10) is already convex and closed (Lemma 2), this exhibits the entire region. □

The binning step in the proof has a simple interpretation. The full codeword index is required by the consumer. At reset, side information identifies the codeword up to a bin of asymptotic size $2^{nI(X;\widehat{X}|S)}$. Only that residual uncertainty is thermodynamically irreducible. The step has also been verified numerically at finite blocklength, by in-bin decoders that succeed from (B, S^n) near the predicted bin rate and fail at every bin rate without S^n.

The work-only endpoint of the region is the minimum Landauer content

$$L(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \widehat{X} \mid S). \tag{15}$$

As an information quantity, $L(D)$ is Steinberg's common-reconstruction rate–distortion function for side information S [7]: the reproduction must be determined without access to S. The operational roles differ. In the common-reconstruction problem, S is decoder side information that must not corrupt reproducibility; here S is held by a reset mechanism that never serves the consumer. The remainder of the paper characterizes $L(D)$, and the region around it, for the source of Section III.

Remark 4 (The Gaussian pair specialization). *From Section III on, the source letter is the jointly Gaussian pair $X = (Y, V)$ introduced there, and the distortion is $d(x, \hat{x}) = (y - \hat{y})^2$ on the read coordinate: the consumer is judged on Y alone, while the encoder observes the pair.*

Remark 5 (Scope of the coding theorem). *Theorem 3 is stated and proved for finite alphabets and bounded per-letter distortion. Its application to the Gaussian source with unbounded quadratic distortion is the standard abstract-alphabet extension of the coding theorem [2]. The Gaussian closed forms of Sections III–VI are evaluations of the single-letter coordinates in* (10)*, and their operational meaning is inherited through that extension.*

III. Pair Sufficiency

Let $X \sim \mathcal{N}(0, \Sigma_x)$ on $\mathbb{R}^d$, i.i.d. across source symbols. The consumer's read is the linear functional $Y = w^\mathsf{T} X$; the third party's copy is a second linear functional observed in noise, $S = \gamma^\mathsf{T} X + U$. Normalize $\mathrm{Var}(w^\mathsf{T} X) = 1$ (rescaling D), write $V = \gamma^\mathsf{T} X / \sigma_\gamma$ with $\rho := \mathbb{E}[YV]$, $|\rho| < 1$, and rescale S (invariantly for all mutual informations) to $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, $\tau^2 > 0$. Write $s = 1 + \tau^2$. Throughout, X, U, and all channel randomness (including any resampling randomness below) are *mutually* independent. This joint form of the independence assumption is load-bearing: it is exactly what the S-kernel argument in the next proof uses.

Reproductions may be taken scalar without loss of optimality: replacing $\widehat{X}$ by $\widehat{Y} := w^\mathsf{T} \widehat{X}$ leaves the distortion unchanged and can only decrease $I(X; \cdot \mid S)$ (conditional data processing), and $\widehat{Y} - X - S$ persists.

Theorem 6 (Pair sufficiency). *In the single-consumer problem $L(D) = \min\{I(X; \widehat{Y} \mid S) : \widehat{Y} - X - S,\ \mathbb{E}(Y - \widehat{Y})^2 \le D\}$, it is without loss of optimality to let the channel depend on X only through $T = (Y, V)$, and then $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. Consequently $L(D)$ depends on $(\Sigma_x, w, \gamma, \tau^2)$, in any ambient dimension d, only through (ρ, τ^2).*

Proof. Given an admissible $p(\hat{y} \mid x)$, draw $\widehat{Y}'$ from the conditional law $p(\hat{y} \mid T = t)$ using fresh randomness. (a) $(T, \widehat{Y}') \stackrel{d}{=} (T, \widehat{Y})$, so the distortion (a functional of $(Y, \widehat{Y})$) is unchanged. (b) $S = V + U$ with U independent of $(X, \widehat{Y})$ and of $(X, \widehat{Y}')$, so in both joints $p(s \mid t, \hat{y}) = p_U(s - v)$: hence $(T, \widehat{Y}', S) \stackrel{d}{=} (T, \widehat{Y}, S)$. (c) Given T, $\widehat{Y}'$ is independent of (X, S), so $I(X; \widehat{Y}' \mid S) = I(T; \widehat{Y}' \mid S) + I(X; \widehat{Y}' \mid T, S) = I(T; \widehat{Y} \mid S) \le I(X; \widehat{Y} \mid S)$, the last step because T is a function of X. (d) $\widehat{Y}' - X - S$ holds since $\widehat{Y}'$ is a function of $(X, \text{fresh noise})$ with the noise independent of (X, U). The same computation applied to a T-channel directly gives $I(X; \widehat{Y} \mid S) = I(T; \widehat{Y} \mid S)$. □

IV. The Function: Scalar Context

One further reduction identifies the class the closed form must beat. *(Marginalized class = scalar corner.)* If the channel depends on X only through Y (that is, $\widehat{Y} - Y - (X,S)$), then given (Y,S), $\widehat{Y}$ is independent of X, so $I(X;\widehat{Y} \mid S) = I(Y;\widehat{Y} \mid S)$, and the class optimum is exactly the scalar-corner value $\frac{1}{2}\log_2\big[(\sigma^2(1-\rho_{YS}^2) + \rho_{YS}^2 D)/D\big]$ of the pair (Y,S) [16], [7]; the corner's converse applies to every $p(\hat{y} \mid y)$, discrete or not.

Proposition 7 (Marginalization is strictly suboptimal). *Let $d=2$, $X \sim \mathcal{N}(0,I_2)$, $Y = X_1$, and $S = X_1 + X_2$ (so $\rho_{YS}^2 = \frac{1}{2}$ and S is X-measurable). For $0 < D < 1$:*

(i) *the marginalized-class optimum is the scalar corner $L_{\mathrm{marg}}(D) = \frac{1}{2}\log_2\frac{1+D}{2D}$;*

(ii) *the vector problem has the exact value*

$$L_{\mathrm{vec}}(D) \;=\; \tfrac{1}{2}\log_2^{+}\frac{1}{2D} \;=\; R_{Y|S}(D),$$

Gray's conditional rate–distortion function of (Y,S) [3] ($\mathrm{Var}(Y \mid S) = \frac{1}{2}$), attained by the linear channel $\widehat{Y} = (1-D)X_1 + DX_2 + N$, $N \sim \mathcal{N}(0, D(1-2D))$, for $D \le \frac{1}{2}$, and by $\widehat{Y} = S/2$ (zero conditional content) for $D \ge \frac{1}{2}$;

(iii) *hence the strict gap $L_{\mathrm{marg}}(D) - L_{\mathrm{vec}}(D) = \frac{1}{2}\log_2(1+D) > 0$ on $(0,\frac{1}{2}]$, and equal to the full marginalized value on $[\frac{1}{2},1)$ (where $L_{\mathrm{vec}} = 0$; the absolute gap is decreasing there, but it is all of L_{marg}).*

Proof. (i) is the reduction above, evaluated at $\sigma^2 = 1$, $\rho_{YS}^2 = \frac{1}{2}$.

(ii) *Converse.* For any admissible $\widehat{Y}$ (Markov $\widehat{Y} - X - S$), $I(X;\widehat{Y} \mid S) \ge I(Y;\widehat{Y} \mid S)$, and the induced conditional channel $p(\hat{y} \mid y,s)$ is feasible for Gray's conditional problem at the same distortion, so $I(Y;\widehat{Y} \mid S) \ge R_{Y|S}(D) = \frac{1}{2}\log_2^{+}\big(\mathrm{Var}(Y \mid S)/D\big) = \frac{1}{2}\log_2^{+}(1/(2D))$. *Achievability.* For $D \le \frac{1}{2}$ take $\widehat{Y} = (1-D)X_1 + DX_2 + N$: distortion $D^2 + D^2 + D(1-2D) = D$; $\mathrm{Var}(\widehat{Y}) = 1-D$, $\mathrm{Cov}(\widehat{Y},S) = 1$, $\mathrm{Var}(\widehat{Y} \mid S) = 1 - D - \frac{1}{2} = \frac{1}{2} - D$, so $I(X;\widehat{Y} \mid S) = \frac{1}{2}\log_2\frac{(1-2D)/2}{D(1-2D)} = \frac{1}{2}\log_2\frac{1}{2D}$. For $D \ge \frac{1}{2}$, $\widehat{Y} = S/2$ is a deterministic function of X, has distortion $\mathbb{E}\big((X_1-X_2)/2\big)^2 = \frac{1}{2} \le D$, and is constant given S, so $I(X;\widehat{Y} \mid S) = 0$.

(iii) Subtract. The closed forms on both sides have been verified numerically. □

V. The Rate–Content Region and Misalignment

Corollary 8 (L is not a functional of the (Y,S) law). *Two instances with the same (Y,S) joint law can have different minimal conditional content: the instances $(\rho^2,\tau^2) = (\frac{1}{2},0)$ and $(\frac{3}{4},\frac{1}{2})$ have identical (Y,S) margins ($\rho_{YS}^2 = \frac{1}{2}$) yet $L = 1.1610$ versus 1.2105 bits at $D = 0.1$, and $L = 0.3685$ versus 0.5228 bits at $D = 0.3$, in both cases strictly below the Steinberg value of the margin (1.2297 and 0.5577 bits respectively). In particular $L(D)$ is not Steinberg's function composed with any modified pair, nor Gray's, nor any functional of the (Y,S) joint law: the encoder's access to the clean context V is load-bearing.*

VI. When the Floor Is Attained

The conditional content of any admissible description is floored by a determinant ratio, and it is natural to ask when the floor is exact. The question is sharpest for a joint description serving two reads, and this section settles it there; the single-read answer appears as a corollary inside the proof. In this section only, $Y = (Y_A, Y_B)$ denotes a pair of normalized reads of the source, jointly Gaussian with the normalized context V, with an arbitrary correlation pattern; the third party again holds $S = V + U$, $U \sim \mathcal{N}(0,\tau^2)$, with X, U, and all channel randomness mutually independent. A joint channel produces $\hat{\mathbf{Y}} = (\widehat{Y}_A, \widehat{Y}_B)$ with $\hat{\mathbf{Y}} - X - S$ and must meet $\mathbb{E}(Y_A - \widehat{Y}_A)^2 \le D_A$ and $\mathbb{E}(Y_B - \widehat{Y}_B)^2 \le D_B$; write $L_{\min}(D_A,D_B) = \min I(X;\hat{\mathbf{Y}} \mid S)$.

Write Σ_Y for the read-pair covariance, $\beta = \mathrm{Cov}(Y,V) \in \mathbb{R}^2$, $\Sigma_{Y|V} = \Sigma_Y - \beta\beta^{\mathsf{T}}$, $\Sigma_{Y|S} = \Sigma_Y - \beta\beta^{\mathsf{T}}/(1+\tau^2)$, and $\Delta = \mathrm{diag}(D_A,D_B) \succ 0$. Every admissible channel obeys the *work floor*

$$L \;\ge\; I(Y;\hat{\mathbf{Y}} \mid S) \;=\; h(Y \mid S) - h(Y \mid \hat{\mathbf{Y}}, S) \;\ge\; \tfrac{1}{2}\log_2\frac{\det\Sigma_{Y|S}}{\det\Delta}, \tag{16}$$

since $h(Y \mid S) = \frac{1}{2}\log_2\big((2\pi e)^2\det\Sigma_{Y|S}\big)$ and $h(Y \mid \hat{\mathbf{Y}}, S) \le \sum_{i\in\{A,B\}} h(Y_i - \widehat{Y}_i) \le \frac{1}{2}\log_2\big((2\pi e)^2 D_A D_B\big)$.

Three facts from the machinery already built carry over. First, pair sufficiency extends verbatim: the resampling proof of Theorem 6 applies with $T = (Y_A, Y_B, V)$, since $S = V + U$ with U jointly independent of

$(X,$ all channel randomness) and the distortions are functionals of the pairs $(Y_i, \widehat{Y}_i)$; so the channel may be restricted to depend on X only through T, and then $I(X; \hat{\mathbf{Y}} \mid S) = I(T; \hat{\mathbf{Y}} \mid S)$. Second, the same mutual independence forces the Markov moment identity $\operatorname{Cov}(S, \hat{\mathbf{Y}}) = \operatorname{Cov}(V, \hat{\mathbf{Y}})$. Third, jointly Gaussian channels $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$, exhaust the problem: for any admissible T-channel with $C = \operatorname{Cov}(T, \hat{\mathbf{Y}})$ and $\Sigma_{\hat{Y}} = \operatorname{Cov}(\hat{\mathbf{Y}})$, set $A := C^{\mathsf{T}} \Sigma_T^{-1}$ and $\Sigma_N := \Sigma_{\hat{Y}} - A\Sigma_T A^{\mathsf{T}} \succeq 0$ (assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping dependent context components); a maximum-entropy bound on these moments gives

$$L \;\geq\; \tfrac{1}{2}\log_2 \frac{\det\bigl(A\Sigma_{T|S}A^{\mathsf{T}} + \Sigma_N\bigr)}{\det \Sigma_N}, \qquad \Sigma_{T|S} = \Sigma_T - \frac{\sigma_{TS}\sigma_{TS}^{\mathsf{T}}}{1+\tau^2}, \quad \sigma_{TS} = \Sigma_T e_V, \tag{17}$$

with equality for the jointly Gaussian channel realizing the same moments, and the distortion constraints $(e_i - A_{i\cdot})^{\mathsf{T}}\Sigma_T(e_i - A_{i\cdot}) + (\Sigma_N)_{ii} \leq D_i$ depend on the channel only through those moments. Consequently $L_{\min}$ is the minimum of (17) over a compact feasible set of pairs (A, Σ_N) and is attained.

A plausible conjecture states the attainment condition for (16) through the third party's conditional covariance $\Sigma_{Y|S}$; the truth involves the encoder-accessible $\Sigma_{Y|V}$, the same S-versus-V substitution as in Proposition 7.

Theorem 9 (Work-floor exactness)**.** *The work floor $\frac{1}{2}\log_2\bigl(\det\Sigma_{Y|S}/\det\Delta\bigr)$ of (16) is attained iff*

$$\text{(a) } \tau^2 = 0 \text{ and } \Delta \preceq \Sigma_{Y|V}, \qquad \text{or} \qquad \text{(b) } \beta = 0 \text{ and } \Delta \preceq \Sigma_Y.$$

In every other case the floor is strict, both on the misaligned branch ($\tau^2 > 0$ and $\beta \neq 0$, at every positive (D_A, D_B)) and outside the semidefinite conditions of (a) and (b). On the misaligned branch with $\Delta \prec \Sigma_{Y|V}$, the deficit obeys the sandwich

$$0 \;<\; L_{\min} - \text{floor} \;\leq\; \tfrac{1}{2}\log_2 \frac{\det\bigl(I - \Sigma_{Y|S}^{-1}\Delta\bigr)}{\det\bigl(I - \Sigma_{Y|V}^{-1}\Delta\bigr)} \;=\; \tfrac{1}{2}\operatorname{tr}\bigl[(\Sigma_{Y|V}^{-1} - \Sigma_{Y|S}^{-1})\Delta\bigr]\log_2 e + O(\|\Delta\|^2) \;\longrightarrow\; 0 \;\; (\Delta \to 0).$$

Proof. Necessity. Suppose the floor is attained. Equality in the chain (16) forces, at the moment level (max-entropy tightness makes the joint WLOG Gaussian): $E_S := \mathbb{E}[\operatorname{Cov}(Y \mid \hat{\mathbf{Y}}, S)] = \Delta$ with per-consumer equality, so $Z := Y - \hat{\mathbf{Y}}$ has $\operatorname{Cov} Z = \Delta$, $\operatorname{Cov}(Z, \hat{\mathbf{Y}}) = 0$, $\operatorname{Cov}(Z, S) = 0$; the latter with the Markov moment identity gives $\operatorname{Cov}(Z, V) = 0$, hence $\operatorname{Cov}(\hat{\mathbf{Y}}, V) = \beta$. Since $E_S \preceq \Sigma_{Y|S}$ always, $\Delta \preceq \Sigma_{Y|S}$ is necessary; at $\tau^2 = 0$ this is $\Delta \preceq \Sigma_{Y|V}$. Tightness of $I(T; \hat{\mathbf{Y}} \mid S) = I(Y; \hat{\mathbf{Y}} \mid S)$ additionally requires $\operatorname{Cov}(V, \hat{\mathbf{Y}} \mid Y, S) = 0$; with the pinned moments this evaluates to $[\Delta \;\; 0]\, M^{-1}\bigl[\beta;\; \operatorname{Cov}(S, V)\bigr] = 0$, $M = \operatorname{Cov}(Y, S)$, which (for $\Delta \succ 0$) holds iff $\beta = 0$ or $\operatorname{Cov}(S, V) = \operatorname{Var} S$, i.e. $\tau^2 = 0$.

Sufficiency (a). At $\tau^2 = 0$ take $Z \sim \mathcal{N}(0, \Delta)$ jointly Gaussian with $\operatorname{Cov}(Z, Y) = \Delta$, $\operatorname{Cov}(Z, V) = 0$; such a Z exists iff $\Delta - \Delta\Sigma_{Y|V}^{-1}\Delta \succeq 0 \Leftrightarrow \Delta \preceq \Sigma_{Y|V}$ (Schur complement). Set $\hat{\mathbf{Y}} = Y - Z$: then $\operatorname{Cov}(\hat{\mathbf{Y}} \mid S) = \Sigma_{Y|S} - \Delta$ and $\operatorname{Cov}(\hat{\mathbf{Y}} \mid T, S) = \Delta - \Delta\Sigma_{Y|V}^{-1}\Delta$, and at $\tau^2 = 0$ (where $\Sigma_{Y|S} = \Sigma_{Y|V} =: \Sigma_c$)

$$L = \tfrac{1}{2}\log_2 \frac{\det(\Sigma_c - \Delta)}{\det\Delta\, \det(I - \Sigma_c^{-1}\Delta)} = \tfrac{1}{2}\log_2 \frac{\det\Sigma_c}{\det\Delta} = \text{floor}.$$

Sufficiency (b). At $\beta = 0$, $\Sigma_{Y|S} = \Sigma_Y$ and a channel depending on X only through Y has $\hat{\mathbf{Y}} \perp S$, so $L = I(Y; \hat{\mathbf{Y}})$, attained at the floor iff $\Delta \preceq \Sigma_Y$ (the exactness condition of the bivariate rate–distortion function under individual distortion constraints [10]); necessity of $\Delta \preceq \Sigma_Y$ is the $E_S \preceq \Sigma_{Y|S} = \Sigma_Y$ bound.

Sandwich. The same Z-construction at $\tau^2 > 0$ is admissible whenever $\Delta \prec \Sigma_{Y|V}$ and yields $L \leq \frac{1}{2}\log_2[\det(\Sigma_{Y|S} - \Delta)/\det(\Delta - \Delta\Sigma_{Y|V}^{-1}\Delta)] = \text{floor} + \frac{1}{2}\log_2[\det(I - \Sigma_{Y|S}^{-1}\Delta)/\det(I - \Sigma_{Y|V}^{-1}\Delta)]$; the ratio is ≥ 1 because $\Sigma_{Y|S} \succeq \Sigma_{Y|V}$ (congruence by $\Delta^{1/2}$ keeps both factors in the PSD order where $\det$ is monotone), and $\to 1$ as $\Delta \to 0$. Strictness (> 0) is the necessity direction, using that $L_{\min}$ is *attained* (compact feasible (A, Σ_N) set, lower semicontinuity); at the non-strict boundary of $\Delta \preceq \Sigma_{Y|V}$ in case (a), where the displayed ratio is $0/0$, attainment follows by $\Delta_t \uparrow \Delta$ feasibility and floor continuity.

The single-read corollary. For one consumer, on the active-floor regime $D \leq \operatorname{Var}(Y \mid V)$, the floor $\frac{1}{2}\log_2(\operatorname{Var}(Y \mid S)/D)$ is attained iff $\rho\tau = 0$: substituting $g_f = (s - \rho^2)/(Ds)$ into the quadratic $P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2)$ of Theorem **??**, whose larger root $g^\star$ gives $L(D) = \frac{1}{2}\log_2 g^\star$, yields $P(g_f) = -\rho^2\tau^2/s$, which is < 0 (hence $g^\star > g_f$ strictly) exactly when $\rho\tau \neq 0$. (The regime qualifier is the single-read shadow of condition (a): for $\tau = 0$, $D > 1 - \rho^2$ the clamped floor and $L = 0$ meet trivially while $g_f < 1$ becomes the quadratic's *smaller* root.) □

VII. Vector Context

VIII. The Binary Solution

The binary case closes completely and, unlike its Gaussian counterpart, with an elementary self-contained proof.

Let (Y, V) be a doubly symmetric binary pair (fair marginals, $\Pr[Y \neq V] = p \in (0, \frac{1}{2})$), context V, side information $S = V \oplus W$ with $W \sim \mathrm{Bern}(q)$, $q \in [0, \frac{1}{2}]$, independent. The record is the reproduction $\hat{Y} \in \{0, 1\}$ itself (richer records only cost more: for any stored Z from which the consumer computes $\hat{Y} = f(Z)$, $I(Y, V; Z \mid S) \geq I(Y, V; \hat{Y} \mid S)$ by the data-processing inequality), Hamming distortion $\Pr[\hat{Y} \neq Y] \leq D$. Write $x * q = x(1-q) + (1-x)q$ and $\ell(x) = \ln \frac{1-x}{x}$.

Theorem 10 (The binary conditional common-reconstruction function)**.** *For $D \in (0, \frac{1}{2})$, the conditional coordinate $L(D) = \min I(Y, V; \hat{Y} \mid S)$ over channels $P(\hat{Y} \mid Y, V)$ (Markov $\hat{Y} - (Y, V) - S$) with $\Pr[\hat{Y} \neq Y] \leq D$ is attained in the two-parameter symmetric family $\hat{Y} = Y \oplus E$, $\Pr[E = 1 \mid Y \oplus V = j] = d_j$, and equals*

$$L(D) \;=\; h_2(u) - (1-p)\, h_2(d_0) - p\, h_2(d_1), \qquad u = a * q, \quad a = (1-p)d_0 + p(1-d_1),$$

where, for $q \in (0, \frac{1}{2}]$ with any $D \in (0, \frac{1}{2})$, and for $q = 0$ with $D \leq p$, the pair (d_0, d_1) solves the active constraint $(1-p)d_0 + pd_1 = D$ together with the tilt equation

$$\ell(d_0) - \ell(d_1) \;=\; 2(1-2q)\, \ell(u).$$

(In the excluded sliver $q = 0$, $D > p$, the constraint is inactive and $L(D) = 0$, attained non-uniquely by records measurable in V alone; the tilt system still has a root there, the degenerate point $d_1 = 1 - d_0$, which evaluates to the correct value 0. For $q > 0$ the constraint is active on all of $(0, \frac{1}{2})$: L is convex, nonincreasing, positive by the last claim below, and 0 at $D = \frac{1}{2}$, hence strictly decreasing.) On the same channel the rate coordinate is $R = 1 - (1-p)h_2(d_0) - p\, h_2(d_1)$ and the discount is exactly $R - L = 1 - h_2(u) = I(\hat{Y}; S)$ (the chain-rule identity $I(Y, V; \hat{Y}) - I(Y, V; \hat{Y} \mid S) = I(\hat{Y}; S)$, valid under the Markov chain, in closed form: $a = \Pr[\hat{Y} \neq V]$, $u = \Pr[\hat{Y} \neq S]$). Anchors: at $q = 0$ the system gives $a = (p - D)/(1 - 2D)$ and $L = h_2(p) - h_2(D)$ (Gray's conditional rate–distortion function [3], for $D \leq p$); at $q = \frac{1}{2}$ it gives $d_0 = d_1 = D$ and $L = 1 - h_2(D)$ (the marginal rate–distortion function; the context is useless exactly when the side information is). For $q \in (0, \frac{1}{2}]$ and $D < \frac{1}{2}$, $L(D) > 0$.

Proof. Reduction and convexity. Binary $\hat{Y}$ is definitional (DPI above). With $P(Y, V \mid S = s)$ fixed and the channel shared across s (Markov), $I(Y, V; \hat{Y} \mid S) = \sum_s P(s)\, I_s$ is a nonnegative combination of mutual informations each convex in the channel, hence convex in the channel; the distortion is linear.

Symmetrization. The global bit flip $\sigma : (y, v, s, \hat{y}) \mapsto (1-y, 1-v, 1-s, 1-\hat{y})$ preserves the joint law of (Y, V, S) and the distortion, and $L(C^\sigma) = L(C)$; averaging C with C^σ preserves distortion and, by convexity, does not increase L. So an optimum exists among σ-symmetric channels, and these are exactly the family $\{(d_0, d_1)\}$: σ-symmetry forces $\Pr[\hat{Y} \neq Y \mid y, v]$ to depend only on $y \oplus v$.

Closed form. On the family, $H(\hat{Y} \mid Y, V) = (1-p)h_2(d_0) + p\, h_2(d_1)$, and $\hat{Y} \oplus S = J \oplus E \oplus W$ $(J = Y \oplus V)$ with $\Pr[J \oplus E = 1] = a$. The step $H(\hat{Y} \mid S) = h_2(a * q)$ needs $\hat{Y} \oplus S \perp S$, which holds because the joint law of $(\hat{Y}, S)$ is σ-invariant (σ-invariant source, σ-symmetric channel), forcing equal conditional flip probabilities given either value of S; the same symmetry gives $H(\hat{Y}) = 1$, hence R.

Tilt equation. Substituting $d_1 = (D - (1-p)d_0)/p$ makes $a = 2(1-p)d_0 + p - D$ affine in d_0; differentiating L and using $h_2'(x) = \ell(x)$ (nats) gives $(1-p)[2(1-2q)\ell(u) - \ell(d_0) + \ell(d_1)] = 0$. Since L restricted to the affine family slice is convex (restriction of a convex function), interior stationarity is sufficient, not merely necessary.

Anchors and positivity. $q = \frac{1}{2}$: $u = \frac{1}{2}$, the tilt equation forces $d_0 = d_1 = D$. $q = 0$: substitution of Gray's backward channel; equality of values is elementary algebra. Positivity, in two branches over the family (which suffices by attainment): if $a \neq \frac{1}{2}$, then $h_2(u) > h_2(a) \geq (1-p)h_2(d_0) + p\, h_2(1-d_1) = (1-p)h_2(d_0) + p\, h_2(d_1)$, where the strict step holds because $u = a * q$ is strictly closer to $\frac{1}{2}$ for $q \in (0, \frac{1}{2}]$ and the second step by concavity; so $L > 0$. If $a = \frac{1}{2}$, then $u = \frac{1}{2}$, $h_2(u) = 1$, and $L = 1 - (1-p)h_2(d_0) - p\, h_2(d_1) \geq 1 - h_2(D) > 0$ by concavity at the distortion. $\square$

Global optimality of the family-plus-tilt solution over all channels, including larger reproduction alphabets, has additionally been verified numerically by unconditional Lagrangian certificates, which assume no convexity.

IX. Discussion

Appendix A
Single-Letterization

For completeness, we justify the last inequalities in (11) and (12). For the rate coordinate,

$$\begin{aligned} I(X^n;\widehat{X}^n) &= H(X^n) - H(X^n \mid \widehat{X}^n) \\ &= \sum_{i=1}^n H(X_i) - \sum_{i=1}^n H(X_i \mid X^{i-1}, \widehat{X}^n) \\ &\geq \sum_{i=1}^n \Big[H(X_i) - H(X_i \mid \widehat{X}_i) \Big] \\ &= \sum_{i=1}^n I(X_i;\widehat{X}_i). \end{aligned} \tag{18}$$

For the work coordinate,

$$\begin{aligned} I(X^n;\widehat{X}^n \mid S^n) &= H(X^n \mid S^n) - H(X^n \mid \widehat{X}^n, S^n) \\ &= \sum_{i=1}^n H(X_i \mid S_i) - \sum_{i=1}^n H(X_i \mid X^{i-1}, \widehat{X}^n, S^n) \\ &\geq \sum_{i=1}^n \Big[H(X_i \mid S_i) - H(X_i \mid \widehat{X}_i, S_i) \Big] \\ &= \sum_{i=1}^n I(X_i;\widehat{X}_i \mid S_i). \end{aligned} \tag{19}$$

Appendix B
Numerical Verification

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